

STS 531: Mathematical Statistics I

Homework V

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1. Show that θ , the acute angle between the two lines of regression is given by

$$\tan(\theta) = \frac{1 - \rho^2}{\rho} \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}.$$

Interpret the cases $\rho = 0$ and $\rho = \pm 1$.

Solution: The equations of the regression lines will be

$$E[Y|X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)$$

and

$$E[X|Y = y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y).$$

The slope of the first regression line is clearly $\rho \frac{\sigma_Y}{\sigma_X}$. Rearranging the second regression equation to

$$(E[X|Y = y] - \mu_X) \frac{\sigma_Y}{\rho \sigma_X} = y - \mu_Y$$

reveals that the slope of this line is $\frac{\sigma_Y}{\rho \sigma_X}$. The angle between the two lines will be given by

$$\theta = \arctan\left(\frac{\sigma_Y}{\rho \sigma_X}\right) - \arctan\left(\rho \frac{\sigma_Y}{\sigma_X}\right).$$

Therefore

$$\tan(\theta) = \tan\left(\arctan\left(\frac{\sigma_y}{\rho\sigma_X}\right) - \arctan\left(\rho\frac{\sigma_Y}{\sigma_X}\right)\right) \quad (1)$$

$$= \frac{\tan\left(\arctan\left(\frac{\sigma_y}{\rho\sigma_X}\right)\right) - \tan\left(\arctan\left(\rho\frac{\sigma_Y}{\sigma_X}\right)\right)}{1 + \tan\left(\arctan\left(\frac{\sigma_y}{\rho\sigma_X}\right)\right)\tan\left(\arctan\left(\rho\frac{\sigma_Y}{\sigma_X}\right)\right)} \quad (2)$$

$$= \frac{\frac{\sigma_Y}{\rho\sigma_X} - \frac{\rho\sigma_Y}{\sigma_X}}{1 + \frac{\sigma_Y^2}{\sigma_X^2}} \quad (3)$$

$$= \frac{\sigma_Y\sigma_X - \rho^2\sigma_X\sigma_Y}{\rho\sigma_X^2 + \rho\sigma_Y^2} \quad (4)$$

$$= \frac{(1 - \rho^2)\sigma_Y\sigma_X}{\rho(\sigma_X^2 + \sigma_Y^2)}. \quad (5)$$

This completes the proof.

For the case of $\rho = 0$, we have $\tan(\theta) = \infty$, which indicates that the lines of regression must be perpendicular. For $\rho = \pm 1$, we will have $\tan(\theta) = 0$, which means the two lines are the same.

2. Let X_1, X_2 and X_3 be uncorrelated random variables having the same standard deviation. Obtain the correlation coefficient between $X_1 + X_2$ and $X_2 + X_3$.

Solution: Let X_1, X_2 and X_3 be σ_1, σ_2 , and σ_3 , respectively. The correlation coefficient between $X_1 + X_2$ and $X_2 + X_3$ is

$$\rho_{X_1+X_2, X_2+X_3} = \frac{\text{Cov}(X_1 + X_2, X_2 + X_3)}{\sigma_{X_1+X_2} \sigma_{X_2+X_3}}. \quad (6)$$

We have

$$\text{Cov}(X_1 + X_2, X_2 + X_3) = E[(X_1 + X_2)(X_2 + X_3)] - E[X_1 + X_2]E[X_2 + X_3] \quad (7)$$

$$= E[X_1X_2 + X_1X_3 + X_2^2 + X_2X_3] - (E[X_1] + E[X_2])(E[X_2] + E[X_3]) \quad (8)$$

$$= E[X_1X_2] + E[X_1X_3] + E[X_2^2] + E[X_2X_3] - E[X_1]E[X_2] - E[X_1]E[X_3] - E^2[X_2] - E[X_2]E[X_3] \quad (9)$$

$$= (E[X_1X_2] - E[X_1]E[X_2]) + (E[X_1X_3] - E[X_1]E[X_3]) + (E[X_2^2] - E^2[X_2]) + (E[X_2X_3] - E[X_2]E[X_3]) \quad (10)$$

$$= \text{Cov}(X_1, X_2) + \text{Cov}(X_1, X_3) + \text{Var}(X_2) + \text{Cov}(X_2, X_3) \quad (11)$$

$$= \sigma_2^2. \quad (12)$$

As for the variance of $X_1 + X_2$, we have

$$\sigma_{X_1+X_2}^2 = E[(X_1 + X_2)^2] - E^2[X_1 + X_2] \quad (13)$$

$$= E[X_1^2 + 2X_1X_2 + X_2^2] - (E[X_1] + E[X_2])^2 \quad (14)$$

$$= E[X_1^2] + 2E[X_1X_2] + E[X_2^2] - E^2[X_1] - 2E[X_1]E[X_2] - E^2[X_2] \quad (15)$$

$$= \sigma_1^2 + \sigma_2^2 + \text{Cov}(X_1, X_2) \quad (16)$$

$$= \sigma_1^2 + \sigma_2^2. \quad (17)$$

Therefore, $\sigma_{X_1+X_2} = \sqrt{\sigma_1^2 + \sigma_2^2}$. By a similar calculation, $\sigma_{X_2+X_3} = \sqrt{\sigma_2^2 + \sigma_3^2}$. Thus, the correlation coefficient between $X_1 + X_2$ and $X_2 + X_3$ is

$$\frac{\sigma_2^2}{\sqrt{(\sigma_1^2 + \sigma_2^2)(\sigma_2^2 + \sigma_3^2)}}.$$

Now, since $\sigma_1 = \sigma_2 = \sigma_3 = \sigma$, this simplifies to

$$\rho_{X_1+X_2, X_2+X_3} = \frac{\sigma^2}{\sqrt{4\sigma^4}} = \frac{1}{2}.$$

3. If X_1 and X_2 are two variables with variances σ_1^2 and σ_2^2 , respectively, and ρ is the correlation coefficient between them, determine k such that $X_1 + kX_2$ and $X_1 + \left(\frac{\sigma_1}{\sigma_2}\right)X_2$ are uncorrelated.

Solution: If $X_1 + kX_2$ and $X_1 + \left(\frac{\sigma_1}{\sigma_2}\right)X_2$ are uncorrelated, then their covariance must be 0. Thus,

$$0 = E \left[(X_1 + kX_2) \left(X_1 + \left(\frac{\sigma_1}{\sigma_2} \right) X_2 \right) \right] - E[X_1 + kX_2]E[X_1 + \left(\frac{\sigma_1}{\sigma_2} \right) X_2] \quad (18)$$

$$= E \left[X_1^2 + \left(\frac{\sigma_1}{\sigma_2} \right) X_1 X_2 + kX_1 X_2 + k \left(\frac{\sigma_1}{\sigma_2} \right) X_2^2 \right] - E^2[X_1] - \frac{\sigma_1}{\sigma_2} E[X_1]E[X_2] - kE[X_2]E[X_1] - k \frac{\sigma_1}{\sigma_2} E^2[X_2] \quad (19)$$

$$= \sigma_1^2 + \frac{\sigma_1}{\sigma_2} \text{Cov}(X_1, X_2) + k \text{Cov}(X_1, X_2) + k \frac{\sigma_1}{\sigma_2} \sigma_2^2 \quad (20)$$

$$= \sigma_1^2 + \frac{\sigma_1}{\sigma_2} \rho \sigma_1 \sigma_2 + k \rho \sigma_1 \sigma_2 + k \frac{\sigma_1}{\sigma_2} \sigma_2^2. \quad (21)$$

Therefore, we must have

$$k(\rho \sigma_1 \sigma_2 + \sigma_1 \sigma_2) = -\sigma_1^2 - \rho \sigma_1^2, \quad (22)$$

so

$$k = \frac{-\sigma_1^2(1 + \rho)}{\sigma_1 \sigma_2(\rho + 1)} \quad (23)$$

$$= \frac{-\sigma_1}{\sigma_2}. \quad (24)$$

4. Suppose both regression curves are, in fact, linear. Specifically, assume that

$$E[Y|X = x] = -\frac{3}{2}x - 2$$

and

$$E[X|Y = y] = -\frac{3}{5}y - 3.$$

(a) Find the correlation coefficient ρ . **(b)** Find $E[X]$ and $E[Y]$.

Solution: **(a)** If the regression curve is linear, then (X, Y) must have a bivariate normal distribution, which means

$$E[Y|X = x] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$$

and

$$E[X|Y = y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y).$$

Therefore, it must be the case that $-\frac{3}{2} = \rho \frac{\sigma_Y}{\sigma_X}$ and $-\frac{3}{5} = \rho \frac{\sigma_X}{\sigma_Y}$. Let $\alpha = \frac{\sigma_Y}{\sigma_X}$. Then we have $-\frac{3}{2} = \rho\alpha$ and $-\frac{3}{5} = \frac{\rho}{\alpha}$. So $\alpha = -\frac{3}{2\rho}$. Thus $-\frac{3}{5} = -\frac{2\rho^2}{3}$. Solving this equation for ρ yields $\rho = \pm \frac{3}{\sqrt{10}}$. The correct solution is $\rho = \frac{3}{\sqrt{10}}$ because the slopes in the regression lines are negative. We can see that this is a valid value for ρ , since $\sqrt{10} > 3$, which makes $0 < \frac{3}{\sqrt{10}} < 1$.

(b) We can find $E[X]$ and $E[Y]$ using the fact that $E[E[X|Y]] = E[X]$ and $E[E[Y|X]] = E[Y]$. Applying this to the first regression line yields

$$\mu_Y = -\frac{3}{2}\mu_X - 2.$$

Applying it to the second regression line yields

$$\mu_X = -\frac{3}{5}\mu_Y - 3.$$

We have two equations in two unknowns. The solution is $\mu_Y = 25$ and $\mu_X = -18$.

5. Suppose X, Y and Z are uncorrelated random variables with standard deviations 5, 12, and 9, respectively. If $U = X + Y$ and $V = Y + Z$, evaluate the correlation coefficient between U and V .

Solution: We can use the general formula found in Problem 2 above. We will have

$$\rho_{X+Y, Y+Z} = \frac{144}{\sqrt{(25 + 144)(144 + 81)}} \quad (25)$$

$$= \frac{144}{\sqrt{169}\sqrt{225}} \quad (26)$$

$$= \frac{144}{13 \times 15} \quad (27)$$

$$= \frac{144}{195} \quad (28)$$

$$\approx 0.7384615. \quad (29)$$

6. The variates, X and Y , have zero means, the same variance σ^2 and zero correlation. Show that

$$U = X \cos(\alpha) + Y \sin(\alpha)$$

and

$$V = X \sin(\alpha) - Y \cos(\alpha)$$

have the same variance σ^2 and zero correlation.

Solution: The variance of U is

$$\text{Var}(U) = E[(X \cos(\alpha) + Y \sin(\alpha))^2] - E^2[X \cos(\alpha) + Y \sin(\alpha)] \quad (30)$$

$$= E[X^2 \cos^2(\alpha) + 2XY \cos(\alpha) \sin(\alpha) + Y^2 \sin^2(\alpha)] - E^2[X] \cos^2(\alpha) - 2 \cos(\alpha) \sin(\alpha) E[X] E[Y] - E^2[Y] \sin^2(\alpha) \quad (31)$$

$$= \cos^2(\alpha) \sigma^2 + 2 \cos(\alpha) \sin(\alpha) \text{Cov}(X, Y) + \sin^2(\alpha) \sigma^2 \quad (32)$$

$$= \sigma^2 (\cos^2(\alpha) + \sin^2(\alpha)) \quad (33)$$

$$= \sigma^2. \quad (34)$$

The variance of V is

$$\text{Var}(V) = E[(X \sin(\alpha) - Y \cos(\alpha))^2] - E^2[X \sin(\alpha) - Y \cos(\alpha)] \quad (35)$$

$$= E[X^2 \sin^2(\alpha) - 2XY \cos(\alpha) \sin(\alpha) + Y^2 \cos^2(\alpha)] - E^2[X] \sin^2(\alpha) + 2 \cos(\alpha) \sin(\alpha) E[X] E[Y] - E^2[Y] \cos^2(\alpha) \quad (36)$$

$$= \sin^2(\alpha) \sigma^2 - 2 \cos(\alpha) \sin(\alpha) \text{Cov}(X, Y) + \cos^2(\alpha) \sigma^2 \quad (37)$$

$$= \sigma^2 (\sin^2(\alpha) + \cos^2(\alpha)) \quad (38)$$

$$= \sigma^2. \quad (39)$$

As for the correlation between U and V , we can show it is zero by showing that the covariance of U and V is zero. We have

$$\text{Cov}(U, V) = E[(X \cos(\alpha) + Y \sin(\alpha))(X \sin(\alpha) - Y \cos(\alpha))] - E[X \cos(\alpha) + Y \sin(\alpha)] E[X \sin(\alpha) - Y \cos(\alpha)] \quad (40)$$

$$= \sigma^2 \cos(\alpha) \sin(\alpha) - \cos^2(\alpha) \text{Cov}(X, Y) + \sin^2(\alpha) \text{Cov}(X, Y) - \sigma^2 \cos(\alpha) \sin(\alpha) \quad (41)$$

$$= 0, \quad (42)$$

since the covariance between X and Y must be 0.

7. Let (X, Y) be a continuous random variable with pdf

$$f(x, y) = \begin{cases} 2, & \text{if } 0 < y < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find $E[Y|X = x]$ and $E[X|Y = y]$.

Solution: The marginal distribution of Y is

$$f_Y(y) = \int_y^1 2 \, dx \quad (43)$$

$$= 2 - 2y, \quad (44)$$

for $0 < y < 1$. The marginal distribution of X is

$$f_X(x) = \int_0^x 2 \, dy \quad (45)$$

$$= 2x, \quad (46)$$

for $0 < x < 1$. Now we can find the conditional pdfs. We have

$$f_{Y|X}(y|x) = \frac{2}{2x} = \frac{1}{x}$$

and

$$f_{X|Y}(x|y) = \frac{2}{2 - 2y}.$$

Using these pdfs, we can find the conditional expectations. We have

$$E[Y|X = x] = \int_0^x \frac{y}{x} \, dy \quad (47)$$

$$= \frac{1}{2x} [y^2]_0^x \quad (48)$$

$$= \frac{x}{2}. \quad (49)$$

The other conditional expectation will be

$$E[X|Y = y] = \int_y^1 \frac{2x}{2 - 2y} \, dx \quad (50)$$

$$= \frac{1}{2 - 2y} [x^2]_y^1 \quad (51)$$

$$= \frac{1 - y^2}{2 - 2y} \quad (52)$$

$$= \frac{(1 - y)(1 + y)}{2(1 - y)} \quad (53)$$

$$= \frac{1 + y}{2}. \quad (54)$$